Alcoholism Recovery Management System

Wachira Christopher Mutie

23/06982

An Information System Project proposal was submitted to KCA Institute in Partial fulfillment of the requirements for the award of the Bachelors in Business Information Technology

KCA Institute

KCA University

Nairobi, Kenya

September 2025

Declaration and Approval

I declare that this project has not been submitted to any other University for the award of a Bachelors of Business Information Technology.

Student Name:

Sign: _____ Date: _____

Supervisor's Name:

Sign: _____

Date: _____

Abstract

A large population of Kenya today struggles with drug and alcohol abuse. Alcohol contributes the most to substance use disorders in Kenya and despite there being a good number of accredited rehabs in the country, not everyone is able to access them. Factors like rehabilitation treatment costs, stigma of addiction, time conflicts and many others keep people from seeking treatment. We aim to create a mobile based solution that will be a recovery resource for recovering alcoholics. Recovering alcoholics need a recovery resource that is free, easily accessible, and can help them keep track of the sober days while connecting them with other recovering addicts. Without the recovery resource it makes it harder for people to be motivated to stay in their recovery journey and maintain a sense of accountability. The proposed solution will enable users to keep track of their sobriety days while interacting with other recovering alcoholics, providing them with motivation.

Table of Contents

Declaration and Approval.....ii

Abstract..... iii

Chapter 1: Introduction.....1

1.1 Background Information.....1

1.2 Problem Statement.....3

1.3 Objectives.....3

1.3.1 General Objectives.....3

1.3.2 Specific Objectives.....4

1.3.3 Research Questions.....4

1.4 Justification.....4

1.5 Scope and Limitations.....5

Chapter 2: Literature Review.....6

2.1 Introduction.....6

2.2 Current Techniques Supporting Alcoholism Treatment.....6

2.2.1 Challenges of Current Technologies Supporting Alcoholism Treatment.....6

2.3 Sobriety Techniques to Boost Alcoholism Recovery.....7

2.3.1 Recovery Based Mobile Applications.....7

2.3.2 Use of Rehabilitation Centers.....7

2.3.3 Peer Support Groups.....8

2.4 Related Works.....8

2.4.1 I Am Sober.....8

2.4.2 24 Hours a Day.....9

2.4.3 Sober Grid..... 11

2.5 Gaps in Related Works.....12

2.6 Conceptual Framework.....12

Table of Figures

I am Sober application (I Am Sober Sobriety App For Android and IOS, n.d.).....9

Example Journal on 24 Hours a Day (Twenty-Four Hours a Day, n.d.).....10

Sober Grid’s UI (Sober Grid - The Worlds Largest Recovery App, n.d.).....11

Conceptual framework.....13

Chapter 1: Introduction

1.1 Background Information

Alcoholism is the inability to control the amounts of alcohol one intakes, with the dependency on alcohol. Alcohol is one of the most abused substances in the world and is still on the rise. Alcohol and substance abuse is a global health concern. Worldwide 3 million deaths are a result of harmful alcohol health abuse which is 5.3% of deaths (World Health Organization, 2019). Of the deaths attributed to alcohol, 28 percent were caused by injuries, 21 percent were caused by digestive disorders and the remainder was due to cardiovascular diseases, cancers, mental disorders, and other health conditions (The United Nations, 2020).

Alcoholism is a prevalent issue that faces a large portion of the Kenyan population either directly or indirectly. The findings of a study undertaken by NACADA in Kenya show that the prevalence of current substance usage among respondents aged 15 – 65 years stands at 12.2% for alcohol; 15.1% of respondents aged 25 - 35 years were currently using alcohol; 5.6% of respondents aged 15 – 24 years were currently using alcohol; and 0.9% of respondents aged 10 – 19 years were currently using alcohol. The percentage of lifetime abstainers (a respondent who has never used any drug or substance of abuse) among respondents aged 15- 65 years was 62.5% in 2017 as compared to 2012 which was 62.9% (NACADA, 2016).

Apart from health disorders alcohol brings greater economic and social losses to individuals and the society at large. In terms of economic losses, the country spends a significant amount of taxpayer money on the treatment of alcoholism and alcohol-related diseases which would have been used in other fields. Alcohol also affects personal relationships, increases sexual violence, domestic violence, homicide, and other criminal activities.

The UN plans to reduce global alcohol consumption in line with target 3.5 of the SDGs, which is to strengthen substance abuse prevention, including narcotic drug abuse and harmful use of alcohol. This can be done by coverage of treatment interventions (pharmacological, psychological, rehabilitation, and aftercare services) for substance use disorders (The United Nations, 2020).

Despite the government policies and interventions alcoholism remains deeply rooted in our society. Although almost all (95 percent) countries have alcohol taxes, fewer than half of them use other price strategies such as banning below-cost selling or volume discounts. In addition, while many countries have bans on advertising on television and radio, the restrictions are rare when it comes to internet and social media advertising.

Alcohol contributes to most of the substance abuse disorders in Kenya despite having several accredited rehabilitation centers in the country. If alcoholism around the world can be cut by a significant percentage it will help us realize SGD number 3, ensure healthy lives, and promote well-being for all at all ages. It will also help in realizing health-related targets including maternal and child health, mental health, infectious diseases, injuries, and poisoning. One of the key sectors of the social pillar of the Kenyan vision 2030 is health and battling alcoholism in our society is a step forward.

The proposed system is a mobile application that serves as an alcoholism recovery resource. Through the application, recovering alcoholics can keep track of the number of days they have been sober, keep a daily online journal to be able to monitor their triggers, and chat with other people who share their struggles.

1.2 Problem Statement

People who are struggling with alcohol addiction require an free recovery resource that can be easily accessed. This mobile app will help them to keep track of the days they are sober. It will also connect them with other people who can relate to their struggles and people to hold them accountable. Without the recovery resource, it would make it harder for people to be self-driven to stay in their recovery journey and maintain a sense of accountability. Alcohol relapse rates vary extensively in clinical studies, but some studies show that people who receive treatment have a short-term remission rate between 20% and 50%. Relapse happens because of the constantly recurring nature of addiction. According to the National Institute on Alcohol Abuse and Alcoholism, proof shows that roughly 90% of people with alcoholism relapse within 4 years after completing treatment. Other studies show that between 20% and 80% of people who receive treatment and experience short-term remission are estimated to relapse in the long term (Mosel, 2021). Without a long-term drug relapse prevention plan, most of the people will not be successful in their attempts to remain sober. A large portion of recovered alcoholics are bound to relapse hence the need for a post-recovery tool that will help them stay in line.

1.3 Objectives

1.3.1 General Objectives

To develop a mobile application that enables recovering alcoholics to keep track of their sobriety and provide a community platform whereby they can speak to someone and help each other in the recovery process.

1.3.2 Specific Objectives

I. To study and analyze the current alcoholism recovery tools.

II. To study and analyze the problems of the current alcoholism recovery tools.

III. To design a mobile application that will be used as an alcoholism recovery tool.

IV. To develop the system.

V. To test and maintain the system.

1.3.3 Research Questions

I. What methods have been used to help alcoholics recover?

II. What tools have been used to help in alcoholism recovery in the past?

III. What are the current problems with current alcoholism recovery technologies?

IV. How will a mobile application that will be used as a recovery tool be designed and developed?

V. How will the proposed solution be tested and maintained?

1.4 Justification

Self-care is one of the methods that is used as a treatment for alcoholism. The system will provide a cheaper platform to help people battle alcohol addiction. The price of rehabilitation depends on the program one chooses, for example, the ninety-day program will cost approximately KSH.390,000 while the forty-five-day program will cost approximately 392,000 (Fountain of hope treatment center, n.d.)

Apart from cost saving, it will save the user time. The alcohol recovery program runs for ninety days or forty-five days continuously (Fountain of hope treatment center, n.d.) The system

will enable users to recover from alcoholism while still doing other things. The system will also seek to help users overcome the stigma that comes with being taken into rehabilitation. The system will also enable the involvement of family members and friends in the journey to sobriety, unlike rehabilitation where there is not much integration with loved ones. The system may also be used as a complement to rehabilitation. It may be used during the time one is in rehabilitation and may also be used as an aftercare tool.

1.5 Scope and Limitations

The system will allow users to register with the platform. Registered users will be allowed to view their sobriety progress and allowed to share their progress with others. The system will not include a feature that will enable the user to talk to a professional on the application.

One of the limitations is that a significant number of rehabs in the country are outdated and fail to adopt new and proven approaches to help people overcome addiction. The industry is slightly flawed and doesn’t always give the best results (Lanzone Morgan,2021). The other limitation is that a large percentage of people do not have smartphones and internet connections. According to the Digital 2021: Kenya report from Data Report only 40% of Kenya's population is using the Internet in 2021. There were only about 21.75 million Internet users in Kenya in January 2021, giving an Internet penetration rate of just 40% (Gilbert, 2021).

Chapter 2: Literature Review

2.1 Introduction

This chapter discusses the literature review and analyzes how recovering alcoholics have been able to access addiction treatments and recovery resources in the past. It also discusses the challenges in the current technologies and related works for people in recovery and those still struggling with alcohol use. It discusses the gaps in previous solutions and modifications for our project.

2.2 Current Techniques Supporting Alcoholism Treatment

There are numerous accredited rehabilitation centers for people struggling with alcohol addiction in the country. Rehabs are the most used recovery resource in the country although not easily accessible and affordable by everyone.

Application developers have combined evidence-based research with technology to come up with recovery-based mobile applications that provide people with addiction treatment and recovery resources. Going for therapy is also a well-known technique used for alcoholism treatment (American Addiction Centers, 2017).

2.2.1 Challenges of Current Technologies Supporting Alcoholism Treatment

Rehabilitation centers are not subject to accountability when they fail to provide treatment that is consistent with medical standards. Moreover, they are flexible to design and implement programs as they choose with little regard to accepted medical standards. Also, a significant number of rehabs in the country are outdated and fail to adopt new and proven

approaches to help people overcome addiction. The industry is slightly flawed and doesn't always give the best results (Lanzone Morgan, n.d.)

A major challenge of recovery-based mobile applications is that a large population lacks smartphones and internet connection and are therefore not able to access them. Also, a lot of people opt to seek help at rehabilitation and therapy centers virtually on a mobile application due to fear of a privacy breach.

2.3 Sobriety Techniques to Boost Alcoholism Recovery

2.3.1 Recovery Based Mobile Applications

To increase access to recovery resources, application developers have come up with recovery-based mobile applications. They make addiction treatment and recovery resources easily accessible to people on their mobile phones. Recovering alcoholics get 24/7 support, can keep track of their sobriety journey, monitor, what triggers them, and connect with people who understand and relate to their struggle (American Addiction Centers, 2017)

This is however faced with several challenges such as lack of access to smartphones by a large part of the population, lack of clinicians' input during the development of the app, and risk of a security breach.

2.3.2 Use of Rehabilitation Centers

A drug rehab center is a place that provides treatment for substance addiction. Treatment may be inpatient or outpatient and short-term or long-term. It generally consists of a personalized blend of therapies for addiction recovery (Vertava Health, n.d.)

Rehab centers are widely used as a substance abuse recovery resource. There are several NACADA approved rehabs countrywide. Rehabilitation centers can however be very expensive especially for the long-term ones as they usually go up to 90 days. It is also time-consuming for the recovering individual and a lot of stigmas come just from being admitted to rehab

2.3.3 Peer Support Groups

Support groups are groups of people who gather to share common problems and experiences associated with a particular problem, condition, illness, or personal circumstance (Paine-Andrews, n.d.) In a support group, members can talk with others who relate to what they are going through. They can share the type of practical insights that can only come from firsthand experience.

In the context of addiction research, positive recovery outcomes are affected by the quality of people's social interactions and perhaps to an even greater extent, by the defining norms of the groups they identify with—that is, using versus recovery groups. One of the challenges faced by peer support groups is a lack of privacy and confidentiality. It is also rather difficult to manage such a group of people.

2.4 Related Works

2.4.1 I Am Sober

I Am Sober is an application that tracks recovery using a sobriety counter to log milestones, tracks how much money the user has saved on drugs or alcohol in their sobriety, and offers motivational quotes for support (The Recovery Village, n.d.) The figure below shows the interface of the mobile application.

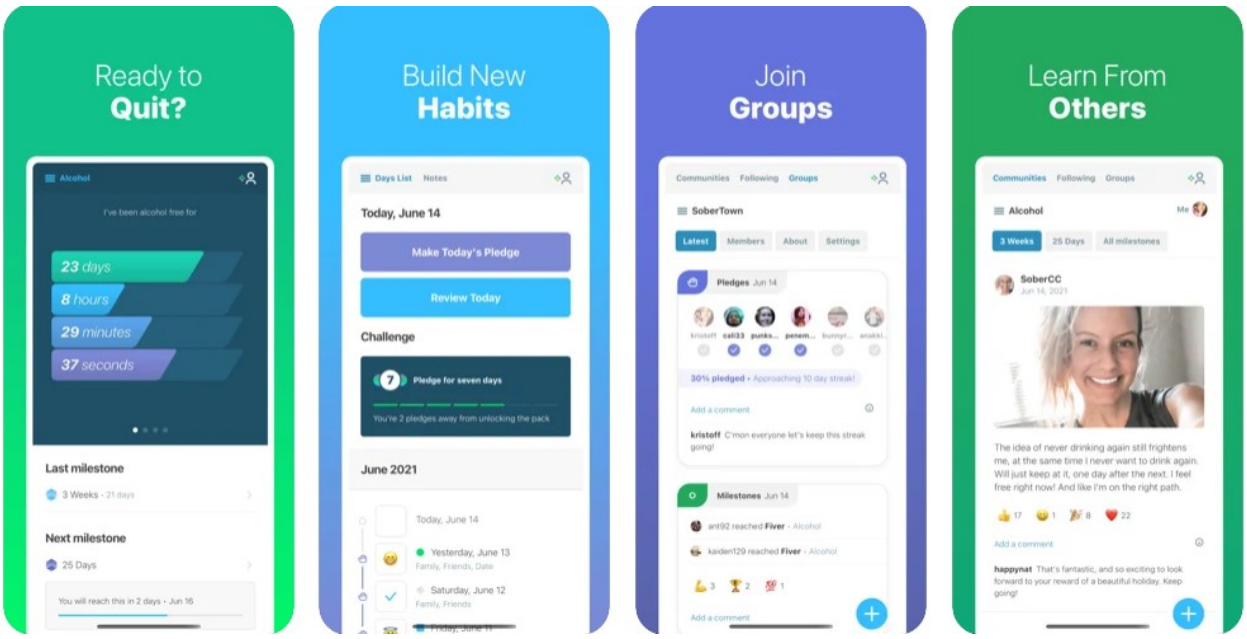

Figure Chapter 2: Literature Review.1 I am Sober application (I Am Sober Sobriety App For Android and IOS, n.d.)

2.4.2 24 Hours a Day

24 Hours a Day is an app that comes with a complete collection of thoughts, meditations, and prayers that make it easy to focus on your sobriety whenever you feel like you could use some support. The daily meditations are inspiring and uplifting. This serves as a great recovery tool and allows one to easily read passages on their phones (American Addiction Centers, 2017).

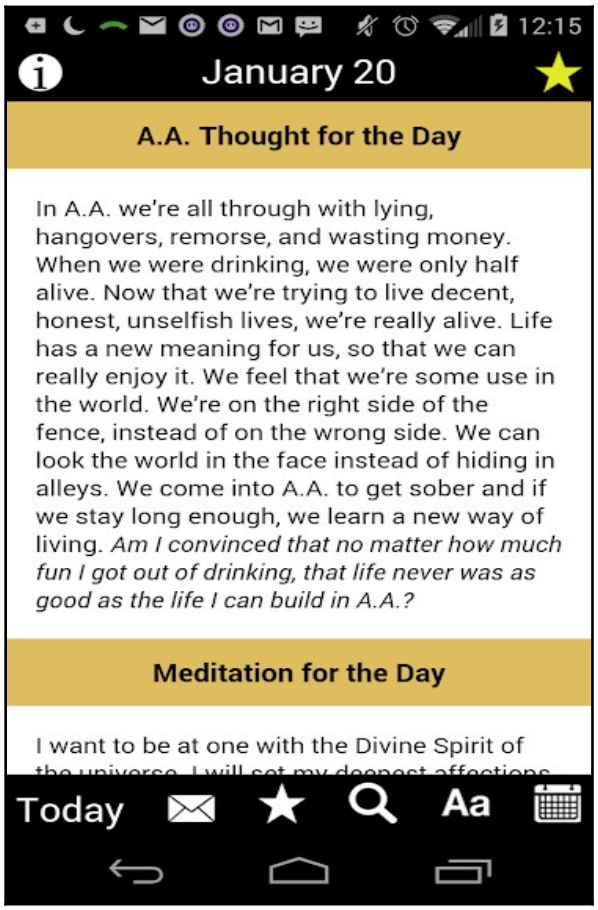

Figure Declaration and Approval.2 Example Journal on 24 Hours a Day (Twenty-Four Hours a Day, n.d.)

2.4.3 Sober Grid

Sober Grid is a social media app that allows you to connect with other sober individuals. You can chat, send private messages, post photographs and videos, and share your interests. The app also includes a sobriety calculator to track the number of sober days you have achieved. Users can remain completely anonymous if they wish (Mental Help, n.d.)

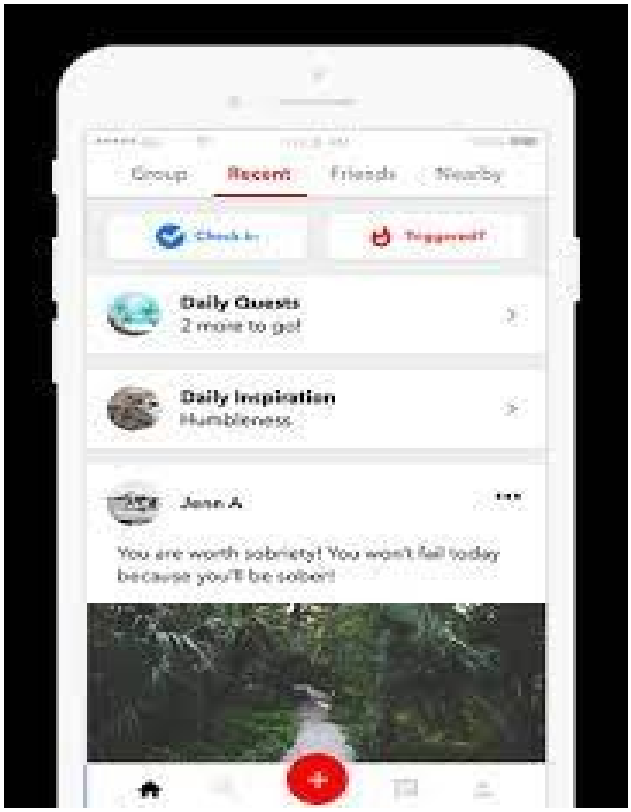

Figure Declaration and Approval.3 Sober Grid's UI (Sober Grid - The Worlds Largest Recovery App, n.d.)

2.5 Gaps in Related Works

Recovery-based mobile applications lack an active medical review. There are several under-developed programs created by amateurs with no intention of providing ongoing support. There needs to be vetting of the applications by clinicians, security experts, and quality assurance software engineers before they are released to the public (Wick & Chiauzzi, 2015, 5)

Related technologies also lack a platform where recovering users can connect and talk to professionals such as counselors. A few modifications of our system are that it enables the user to keep track of the number of days they have been sober and track their savings from overcoming their addictions.

2.6 Conceptual Framework

The person who wishes to use our application registers their account then proceeds to log into the system. After a successful login, the user can build their profile by filling out their personal information. They specify the date they began their sobriety for the system to calculate how long they have been sober. The user can log their emotional status and chat with other users on the community tab.

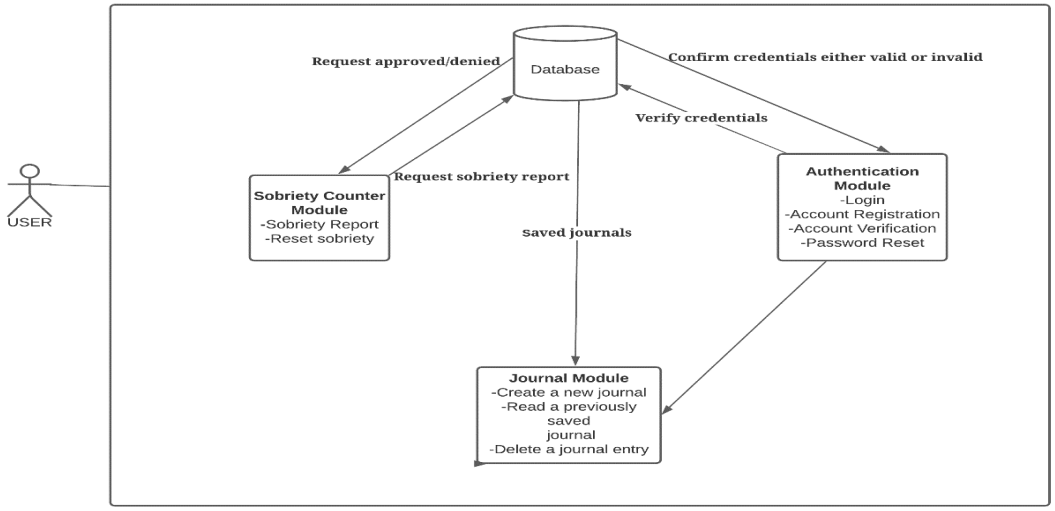

Figure Chapter 2: Literature Review.4 Conceptual framework